

☒ 40 MCQs with Answers and Explanations

☒ EASY (10 Questions)

Q1. What is a Python library?

- A. A database
- B. A file reader
- C. Reusable set of functions
- D. A user input method

Answer: C

Explanation: A library contains pre-written reusable code.

Q2. What is the purpose of map() in Python?

- A. Filters items
- B. Maps file locations
- C. Applies a function to all items
- D. Finds index

Answer: C

Explanation: map() applies a function to each item.

Q3. What does the lambda keyword do?

- A. Starts a loop
- B. Declares a class
- C. Defines an anonymous function
- D. Initializes a list

Answer: C

Explanation: lambda creates anonymous functions.

Q4. What is a server in networking?

- A. A user program
- B. A computer that provides services
- C. A JavaScript library
- D. A compiler

Answer: B

Explanation: A server handles and responds to client requests.

Q5. Which of these is a web server?

- A. Python
- B. Django
- C. Apache
- D. Flask

Answer: C

Explanation: Apache is a web server.

Q6. Which protocol does a web server use?

- A. FTP
- B. SMTP
- C. HTTP
- D. SSH

Answer: C

Explanation: Web servers use HTTP or HTTPS.

Q7. Which command starts a Python web server?

- A. web.run()
- B. python -m http.server
- C. startweb()
- D. http.run()

Answer: B

Explanation: Starts a simple HTTP server.

Q8. What does filter() do?

- A. Copies items
- B. Deletes values
- C. Filters data using a function
- D. Splits lists

Answer: C

Explanation: Filters based on a condition.

Q9. Which of the following is client-side scripting?

- A. Python
- B. PHP
- C. JavaScript
- D. Flask

Answer: C

Explanation: JavaScript runs in browsers.

Q10. Which function reduces a list to a single value?

- A. map()
- B. filter()
- C. reduce()
- D. lambda()

Answer: C

Explanation: reduce() performs rolling computation.

☒ **MEDIUM (20 Questions)**

Q11. What does import math do?

- A. Loads math functions
- B. Opens calculator
- C. Imports numeric values
- D. Adds decimals

Answer: A

Explanation: Provides access to math library functions.

Q12. Which functional tool returns only truthy results?

- A. map()
- B. reduce()
- C. filter()
- D. lambda()

Answer: C

Explanation: filter() filters values that return True.

Q13. What does a client do in client-server architecture?

- A. Serves resources
- B. Requests services
- C. Updates server code
- D. Hosts websites

Answer: B

Explanation: The client initiates requests.

Q14. Flask is a:

- A. Frontend language
- B. Web server
- C. Python web framework
- D. Database

Answer: C

Explanation: Flask is used for building web apps.

Q15. `reduce(lambda x, y: x+y, [1,2,3])` returns:

- A. 1
- B. 3
- C. 6
- D. Error

Answer: C

Explanation: Adds values $1+2+3 = 6$.

Q16. `lambda x: x*2` is an example of:

- A. Decorator
- B. Loop
- C. Anonymous function
- D. Method

Answer: C

Explanation: lambda creates unnamed functions.

Q17. Which of these uses a request-response model?

- A. TCP
- B. SMTP
- C. HTTP
- D. UDP

Answer: C

Explanation: HTTP is based on request-response.

Q18. What does the Flask route `@app.route('/')` define?

- A. The database
- B. The homepage URL
- C. A loop
- D. A filter

Answer: B

Explanation: Maps `/` to a view function.

Q19. A client connects to a server using:

- A. Signal
- B. TCP/IP
- C. Filesystem
- D. API only

Answer: B

Explanation: TCP/IP protocols are used.

Q20. Which of these are Python third-party libraries?

- A. os
- B. math
- C. numpy
- D. print

Answer: C

Explanation: numpy is installed via pip.

Q21. What port does Python's HTTP server use by default?

- A. 80
- B. 443
- C. 8000
- D. 3306

Answer: C

Explanation: Port 8000 is default for Python HTTP server.

Q22. In which paradigm is immutability preferred?

- A. Object-oriented
- B. Procedural
- C. Functional
- D. Structured

Answer: C

Explanation: Functional programming favors immutability.

Q23. Which layer does HTTP work on?

- A. Transport
- B. Network
- C. Application
- D. Link

Answer: C

Explanation: HTTP is an application layer protocol.

Q24. Which of these is not a client-side scripting language?

- A. JavaScript
- B. HTML
- C. Python
- D. CSS

Answer: C

Explanation: Python is server-side.

Q25. Which library is used for HTTP requests in Python?

- A. http
- B. socket
- C. requests

D. json

Answer: C

Explanation: requests is used to send HTTP requests.

Q26. What is a web server's role?

A. Run background scripts

B. Serve files over HTTP

C. Compile programs

D. Run shell scripts

Answer: B

Explanation: It delivers web content to clients.

Q27. Which is the correct way to declare a lambda function?

A. lambda(x) => x*2

B. lambda x: x*2

C. function x() x*2

D. lambda(x): return x*2

Answer: B

Explanation: lambda x: x*2 is correct syntax.

Q28. Which server is best for static files?

A. Django

- B. Flask
- C. Apache
- D. SQLite

Answer: C

Explanation: Apache is efficient for static content.

Q29. `map(lambda x: x+3, [1,2])` returns:

- A. [1, 2]
- B. [3, 5]
- C. [4, 5]
- D. [3, 4]

Answer: C

Explanation: Adds 3 to each item.

Q30. In client-server model, what handles database?

- A. Client
- B. Server
- C. Router
- D. Browser

Answer: B

Explanation: Server typically manages database access.

☒ HARD (10 Questions)

Q31. What is `functools.reduce()` used for?

- A. Accumulates values to single result
- B. Filters based on condition
- C. Applies function to each item
- D. Splits a string

Answer: A

Explanation: `reduce()` combines all elements.

Q32. Which of these protocols is **stateless**?

- A. FTP
- B. HTTP
- C. SSH
- D. Telnet

Answer: B

Explanation: HTTP doesn't retain connection state.

Q33. Which error occurs if socket port is already in use?

- A. `TypeError`
- B. `AddressAlreadyInUseError`
- C. `OSError`

D. ValueError

Answer: C

Explanation: OS throws error if port is bound.

Q34. Python's `socket.send()` requires:

A. String

B. Integer

C. Bytes

D. JSON

Answer: C

Explanation: You must send bytes over sockets.

Q35. Which Python library helps build RESTful APIs quickly?

A. os

B. http

C. Flask

D. FastAPI

Answer: D

Explanation: FastAPI is designed for APIs.

Q36. Which tool helps inspect HTTP traffic between client and server?

A. Wireshark

- B. MySQL
- C. pip
- D. JSON

Answer: A

Explanation: Wireshark captures packets.

Q37. What does the term “stateless protocol” mean?

- A. Doesn't require memory
- B. Stores data on server
- C. Treats each request as independent
- D. Logs data automatically

Answer: C

Explanation: No memory of previous interactions.

Q38. Which function in Flask handles incoming URL?

- A. @flask.route()
- B. @app.get()
- C. @app.route()
- D. url_map()

Answer: C

Explanation: Maps URL to handler function.

Q39. What does `socket.listen(5)` do?

- A. Sends 5 messages
- B. Listens on port 5
- C. Queues 5 connections
- D. Listens 5 times

Answer: C

Explanation: 5 is max queue for incoming connections.

Q40. In REST, GET method is:

- A. To update resource
- B. To delete data
- C. To retrieve data
- D. To restart server

Answer: C

Explanation: GET fetches resources.